Clubs and Societies at BIT Jaipur Campus

Various student clubs and societies are created at BIT Jaipur Campus to help the students to learn and grow in all aspects of professional life. The various clubs and societies are:

1. **NSS (National Service Scheme):**
   The National Service Scheme (NSS) at Birla Institute of Technology, Off-Campus Jaipur, embodies the spirit of selfless service with its slogan, "Not Me, But You". Founded in the campus, in September 2023, the organization fosters social responsibility, leadership, and teamwork among students. Since its inception, NSS BIT Jaipur has successfully conducted over three hundred activities, making a significant impact both on and off campus.

   In the most recent semester (SP-2025), first and second-year NSS volunteers collectively conducted over one hundred initiatives. From impact driven project to International Women's Day, these activities span diverse areas such as community service, environmental conservation, awareness campaigns, and skill development, reflecting the vibrant culture of service and learning within the institution. NSS BIT Jaipur continues to be a catalyst for change, empowering students to contribute meaningfully to society.

   NSS BT Jaipur not only focuses on community welfare but also emphasizes the holistic development of its volunteers. Through participation in various events, students develop essential life skills such as teamwork, empathy, and effective communication. The organization also provides a platform for students to highlight their leadership potential by organizing and leading activities, fostering a sense of accountability and commitment. By integrating social service with academics, NSS BIT Jaipur nurtures a generation of socially aware and responsible individuals who are not just contributors to the community but also torchbearers of positive change for the future.

   In conclusion, NSS BIT, Jaipur serves as a beacon of selflessness and dedication, instilling values that extend beyond the boundaries of the campus. By addressing pressing societal issues and inspiring students to actively participate in creating solutions, the organization exemplifies the transformative power of youth engagement. With a steadfast commitment to service, NSS BIT Jaipur continues to empower its volunteers, leaving a lasting impact on the community and shaping a brighter, more inclusive future for all.

NSS Program Officers:

- Mr. Deepak Chaturvedi
- Dr. Puneet Sharma

Faculty Coordinators:

- Dr. Rajni Agarwal
- Dr. Ravindra Kumar

N.S.S. Unit Lead:

- Priyanshu Shekhar (B. Tech CSE)

2. **Cultural Club:**

Cultural Club – BIT Jaipur Campus

The **Cultural Club** is one of the most active and celebrated student clubs at **BIT Jaipur Campus**. It promotes diversity, creativity, collaboration, and cultural awareness among students. The club provides a platform for students to discover, develop, and display their talents through various artistic, literary, and social activities. The club regularly organizes events, performances, awareness campaigns, competitions, and cultural programs throughout the academic year.

Leadership

Faculty In-Charge: Dr. Rajni Agarwal

Student Coordinator: Prabal Goyal (ECE VI)

Constituent Clubs

The Cultural Club comprises five major sub-clubs:

1. Music Club – "Sonance"

Faculty Coordinators

- Dr. G. S. Malhotra
- Mr. Puneet Sharma

Student Coordinators

- Suhani Mathur (ECE VI)
- Tanushree Paul (ECE IV)

About the Club

Sonance is the official music club of BIT Jaipur. It provides a platform for students interested in music to learn, perform, collaborate, and experiment with various musical genres. The club promotes musical expression, teamwork, and creativity while encouraging students to improve their vocal and instrumental skills. It actively participates in college events, annual fests, alumni meet, and national celebrations.

Activities

- Live musical performances
- Band performances
- Cultural fest participation
- Independence Day celebrations
- Alumni Meet performances.
- Music practice sessions and jam sessions

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2. Girl Up – "Anira"

Faculty Coordinators

- Dr. Seema Gaur
- Dr. Kusum Lata Mishra

Student Coordinators

- Vrinda Malik (CSE VI)
- Charu S. Gupta (ECE VI)

About the Club

Girl Up Anira is a chapter of the global **Girl Up Movement**, founded by the United Nations Foundation in 2010. The club focuses on empowering young women through leadership development, awareness campaigns, and community engagement initiatives. Its activities revolve around five major themes:

- Education
- Health
- Safety
- Leadership
- Rights

The club conducts awareness campaigns, workshops, discussions, fundraising activities, and "Safe Space" sessions to encourage gender equality and women's empowerment.

Activities

- Safe Space discussions
- Women empowerment campaigns
- Awareness drives
- Leadership development programs
- Social outreach initiatives

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3. Drama Club – "Irtiqa"

Faculty Coordinators

- Dr. Seema Gaur
- Mr. Gautam Goswami

Student Coordinators

- Satvik Sharma (BCA VI)

- Vrinda Malik (CSE VI)

About the Club

Irtiqa is the official drama and theatre club of BIT Jaipur. The club brings together students interested in acting, script writing, directing, stage management, and performing arts. It aims to nurture creativity, confidence, communication skills, and artistic expression through theatrical performances and street plays.

Activities

- Stage plays
- Nukkad Natak (street plays)
- Independence Day performances
- Republic Day performances
- Theatre workshops
- Script writing and direction

Skills Developed

- Public speaking
- Acting
- Stage presence
- Teamwork
- Leadership
- Creative storytelling

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4. Literature Club – "Geek Retreat"

Faculty Coordinators

- Dr. Ritu Pareek
- Dr. G. S. Malhotra

Student Coordinators

- Ms. Surmai Sharma (MBA II)
- Ms. Purviya Gupta (BBA VI)

About the Club

Geek Retreat serves as a platform for students interested in literature, language, writing, debating, reading, public speaking, and critical thinking. The club encourages intellectual growth by organizing activities that develop communication and analytical skills.

Activities

- Debates
- Creative writing competitions
- Poetry sessions
- Book discussions
- Public speaking events
- Presentation competitions
- Literary workshops

Skills Developed

- Communication
- Writing
- Presentation
- Critical thinking
- Analytical reasoning
- Public speaking

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5. Dance Club – "Nrityarizz"

Faculty Coordinators

- Dr. Kusum Lata Mishra
- Mrs. Jyoti Sharma

Student Coordinators

- Anshika Nigam (ECE VI)
- Mahi Jain (ECE VI)

About the Club

Nrityarizz is the official dance club of BIT Jaipur. The club provides opportunities for students enthusiastic about various dance forms to learn, practice, perform, and compete. The club regularly participates in cultural festivals and institutional events, highlighting talent through solo, duet, and group performances.

Activities

- Cultural fest performances
- Group dance competitions.
- Solo and duet performances
- Choreography workshops
- Festival celebrations
- Stage performances

Skills Developed

- Team coordination
- Stage confidence
- Choreography
- Physical fitness
- Artistic expression

Major Objectives of the Cultural Club

- Promote cultural diversity and inclusiveness.
- Provide a platform for artistic and creative expression.
- Develop leadership, teamwork, and communication skills.
- Encourage participation in extracurricular activities.
- Preserve and celebrate Indian cultural heritage.

- Foster social awareness and community engagement.
- Organize cultural events, competitions, workshops, and performances.
- Help students discover and highlight hidden talents.

Benefits of Joining the Cultural Club

- Personality development
- Networking and collaboration
- Event management experience
- Leadership opportunities
- Public speaking and communication skills
- Confidence building
- Artistic and creative growth
- Exposure to inter-college and college-level cultural events

Overall, the Cultural Club serves as the umbrella organization for the **Music Club (Sonance), Girl Up Anira, Drama Club (Irtiqa), Literature Club (Geek Retreat), and Dance Club (Nrityarizz)**, making it one of the primary centres for student engagement and talent development at BIT Jaipur.

3. **<u>Sports Club:</u>**

The BIT Jaipur Sports Club, officially known as **Cavorts**, is the central student-driven sports committee at the Birla Institute of Technology (BIT) Jaipur Campus. It manages all physical fitness activities, sports events, and intramural tournaments throughout the academic year.

Core Activities & Facilities

- **Annual Sports Fest ("Cavorts"):** The club organizes this major inter-collegiate sports event featuring tournaments in cricket, basketball, volleyball, badminton, and athletics.
- **Sangram:** A high-intensity, multi-day sports event hosted within the intramural calendar for students.
- **Campus Facilities:** The club coordinates student access to on-campus sporting infrastructure, including a well-equipped gymnasium, basketball courts, and expansive grounds for cricket and football.

Club Leadership

- **Faculty Advisors:** Dr. Piyush Gupta and Mr. Vikas Sangvan serve as the primary coordinators and mentors for the sports club.

4. **ProgramiX Club:**

ProgramiX Club is the official programming club of the Department of Computer Science & Engineering at BIT Jaipur Campus. The club aims to enhance coding skills, promote innovation, and foster collaboration among students interested in programming and technology. It provides a platform for students to learn, compete, and grow through coding competitions, hackathons, workshops, mentorship programs, and networking opportunities with peers and industry professionals.

Faculty Coordinators

- Dr. Madan Mohan
- Dr. Pankaj Gupta

Core Team

- **Charchit Jain** – Treasurer & Management Lead
- **Vansh Jain** – Social Lead & DSA Co-Lead
- **Aditya Goyal** – DSA Lead & Java Lead
- **Aviral Sharma** – Data Science Co-Lead & Python Co-Lead
- **Mehak Jha** – Data Science Lead & Web Development Co-Lead
- **Karishma Jethwani** – Web Development Lead & Java Co-Lead
- **Shreya Verma** – Python Lead

Major Activities

- Coding Competitions
- Hackathons
- Technical Workshops
- Mentorship Programs
- Networking Sessions
- Skill Development in:
    - Data Structures & Algorithms (DSA)
    - Java
    - Python
    - Web Development
    - Data Science

Recent Highlight

The **ProgramiX Club Induction Program** was held on **13 February 2025** to introduce new members to the club's vision, activities, and growth opportunities. The event featured addresses by **Dr. Shripal Vijayvargiya (HOD, CSE)** and mentors **Dr. Madan Mohan** and **Dr. Pankaj Gupta**, who encouraged students to actively participate in coding, innovation, and continuous learning.

Objectives

- Improve coding and critical thinking skills.
- Encourage participation in competitive programming and hackathons.
- Provide mentorship and peer learning opportunities.

- Build expertise in emerging technology domains.
- Create a collaborative and supportive technical community.

5. **AI Club:**

   The **Artificial Intelligence Club** is a research-driven and innovation-focused student community at BIT Jaipur Campus that promotes learning, research, and practical implementation of Artificial Intelligence technologies. The club provides students with opportunities to work on AI projects, explore emerging technologies, and develop solutions for real-world industrial and societal challenges.

   Faculty Mentors

   - Dr. Vishambhar Pathak (Assistant Professor, CSE)
   - Dr. Smitha Jha (Assistant Professor, CSE)

   Student Coordinators

   - Aryan Gaur (CSE)
   - Meher Soni (CSE)

   Focus Areas

   The club works across several AI domains, including:

   - Machine Learning & Deep Learning
   - Natural Language Processing (NLP)
   - Computer Vision
   - Generative AI
   - AI in Cybersecurity
   - AI for Healthcare & Well-being
   - Robotics & Autonomous Systems
   - AI Ethics & Responsible AI
   - AI in Finance & Business Intelligence

   Application Areas

   - Healthcare & Biomedical Research
   - Finance & Banking
   - Smart Cities & IoT
   - Autonomous Vehicles
   - Retail & E-Commerce
   - Education & Personalized Learning

   Major Activities

   - AI Workshops and Hands-on Training
   - Hackathons and Coding Competitions
   - Technical Paper Writing and Presentations
   - Project Development and Research Collaborations
   - Guest Lectures and Webinars

- Group Discussions and Quizzes
- Industry and Research Lab Visits
- AI Startup and Entrepreneurship Support

Career Opportunities Promoted

The club helps students prepare for careers in:

- Artificial Intelligence Research
- Data Science and Analytics
- Machine Learning Engineering
- AI-driven Cybersecurity
- Robotics and Automation

Students gain exposure to skills relevant for organizations such as Google, Microsoft, IBM, ISRO, DRDO, Tesla, and AI-based startups.

Objective

Bridge the gap between academic learning and industry requirements by providing practical AI knowledge, research opportunities, technical mentorship, and industry-oriented skill development.

6. **<u>Electronika Club:</u>**

About the Club

Electronika Club is the official technical club of the Department of Electronics & Communication Engineering at BIT Jaipur Campus. The club aims to develop technical competence, innovation, and critical thinking skills among students by providing opportunities for practical learning, research, project development, and industry-oriented activities. The club encourages students to bridge the gap between theoretical knowledge and real-world applications through hands-on experience and technological exploration.

The mission of the club is to build a complete technological persona by promoting knowledge sharing, practical implementation of concepts, and innovation for societal benefit. The club prepares students for future industrial and technological challenges while fostering an application-oriented mindset.

Faculty Mentors

- Mr. Snehanshu Shekhar (Assistant Professor, ECE)
- Mr. Gaurav Jain (Assistant Professor, ECE)

Student Coordinators

- Harshit Mishra (ECE, Final Year)
- Vaidushi Sharma (ECE, Final Year)

Major Technical Domains

1. Robotics, Microprocessors, Microcontrollers and Applied Instrumentation

This domain focuses on automation, intelligent control systems, embedded computing, data acquisition, sensor networks, telemetry, and AI-based robotic

systems. Students gain exposure to modern robotic technologies and practical applications of microprocessors and microcontrollers in industrial and societal systems.

Key Areas:

- Robotics and Automation
- Sensor Networks
- Telemetry Systems
- Data Acquisition Systems
- SCADA Systems
- Artificial Intelligence Integration
- Machine Learning Applications
- Natural Language Processing
- Embedded Control Systems

The domain encourages students to understand and develop advanced robotic systems inspired by technologies such as Sophia, Atlas, Ameca, Nadine, and ASIMO.

2. Signal Processing

Signal Processing forms the foundation of modern engineering applications involving data analysis and communication systems. This domain focuses on the processing, conditioning, analysis, and interpretation of analog and digital signals.

Key Areas:

- Digital Signal Processing (DSP)
- Image Processing
- Communication Systems
- Audio and Speech Processing
- Sensor Signal Analysis
- Signal Conditioning
- Data Analytics

Students develop practical skills required for processing and analysing signals used in communication, aerospace, industrial, and research applications.

3. Microelectronics and Embedded Systems

This domain focuses on the design, fabrication, and implementation of integrated circuits and embedded systems that power modern electronic devices.

Key Areas:

- Integrated Circuit Design
- VLSI and ULSI Technologies
- System-on-Chip (SoC)

- Embedded Systems
- MEMS and NEMS Technologies
- Semiconductor Devices
- Chip Design and Testing

Students gain insights into advanced microelectronic technologies that drive modern computing, communication, and intelligent systems.

Application Areas

The technologies explored under Electronika Club have applications in:

- Automation and Industrial Control
- Process Industries
- Smart Healthcare Systems
- Smart Agriculture
- Transportation Systems
- Environmental Monitoring
- Industrial Internet of Things (IIoT)
- Smart Energy Distribution and Metering
- Smart Vehicles
- Cloud-Based Resource Sharing
- AI-Based Robotic Systems
- VLSI and Semiconductor Industry
- Chip Design and Testing

Objectives of the Club

- Promote practical and application-based learning.
- Encourage innovation and technological creativity.
- Develop industry-relevant technical skills.
- Foster research and development culture among students.
- Prepare students for future technological and industrial requirements.
- Build self-sustainable and industry-ready professionals.

General Activities

The club organizes a variety of technical and professional development activities, including:

- Study of technical magazines, journals, and research publications.
- Design, modelling, simulation, fabrication, and evaluation of engineering systems.
- Development of technical projects and prototypes.
- Technical paper writing and presentations.
- Quizzes and group discussions.
- Expert lectures by industry professionals and academicians.

- Workshops and hands-on training sessions.
- Interaction with industrial and research experts.
- Industrial visits and institutional tours.
- Participation in research and development projects.

Benefits of Joining Electronika Club

Students associated with Electronika Club gain:

- Practical exposure to emerging technologies.
- Hands-on project development experience.
- Research and innovation opportunities.
- Industry-oriented technical skills.
- Teamwork and leadership experience.
- Enhanced problem-solving and analytical abilities.
- Better preparation for higher studies, research careers, and industry roles.

Electronika Club serves as a platform for aspiring engineers to transform ideas into practical solutions while contributing to technological advancement and societal development.

7. **Electrical & Electronics Engineering (EEE) Technical Clubs**

The Department of Electrical & Electronics Engineering at BIT Jaipur Campus promotes technical learning and innovation through specialized clubs that provide students with practical exposure, industry-oriented knowledge, and opportunities for research and project development.

1. Power Electronics and Devices Club

About the Club

The **Power Electronics and Devices Club** focus on the study and application of power electronic systems used across industries such as manufacturing, automation, transportation, computing, renewable energy, and electric vehicles. Power electronics plays a crucial role in efficient energy conversion, motor drives, power quality improvement, smart grids, renewable energy integration, and pollution reduction. With the growing adoption of smart power devices and electric vehicles, this field has become increasingly significant in modern engineering.

Key Focus Areas

- Power Electronics Devices (SCRs, MOSFETs, IGBTs)
- Industrial Electronics and Drives
- Active Power Factor Correction
- Harmonic Compensation
- Renewable Energy Integration

- Smart Power Devices
- Electric and Hybrid Electric Vehicles
- Control Circuits and Power Converters

Activities

- Student seminars on emerging trends in power electronics.
- Expert lectures by industry professionals and academicians.
- In-house technical quizzes.
- Awareness programs, poster competitions, and slogan competitions.
- Design and fabrication of circuits using SCRs, MOSFETs, IGBTs, and other power devices.
- Development of small control circuits.
- Projects involving renewable energy integration and electric vehicle technologies.

Benefits

- Hands-on experience with power electronic components.
- Understanding of industrial automation and energy systems.
- Exposure to electric vehicle technologies.
- Development of practical design and implementation skills.

2. Alternative Sources of Energy Club

About the Club

The **Alternative Sources of Energy Club** focuses on renewable and sustainable energy technologies. The club promotes awareness, research, and practical learning related to solar energy, wind energy, energy conservation, smart grids, and clean energy solutions.

The club addresses the growing global demand for environmentally friendly energy resources and supports the transition toward sustainable power generation and energy management systems.

Key Focus Areas

- Solar Energy Systems
- Wind Energy Systems
- Renewable Energy Integration
- Energy Conservation
- Smart Grids
- Distributed Energy Resources
- Microgrids
- Electric Vehicle Integration
- Sustainable Energy Management

Objectives

- Promote the use of clean and renewable energy sources.
- Increase awareness about energy conservation.

- Support sustainable development and environmental protection.
- Encourage innovation in renewable energy technologies.
- Develop technical understanding of modern energy systems.

Activities

- Student seminars on recent developments in the energy sector.
- Guest lectures by experts in renewable energy and energy management.
- Technical quizzes and competitions.
- Energy awareness campaigns.
- Distribution of informational leaflets in colleges and nearby schools.
- Painting and poster competitions on energy conservation themes.
- Discussions on renewable energy technologies and their applications.

Benefits

- Knowledge of renewable energy technologies.
- Understanding of energy management and sustainability.
- Exposure to modern power systems and smart grid concepts.
- Opportunities to contribute to environmental awareness initiatives.

Overall Objectives of EEE Clubs

The EEE technical clubs aim to:

- Enhance practical and application-oriented learning.
- Promote innovation and technical creativity.
- Develop industry-relevant skills.
- Encourage research and project development.
- Create awareness about sustainable technologies.
- Prepare students for careers in power systems, renewable energy, electric vehicles, automation, and industrial electronics.

Together, these clubs provide students with a platform to apply classroom knowledge to real-world engineering challenges while preparing them for future technological advancements and industry requirements.

8. **Galvanizers Club**

The **Galvanizers Club** is the flagship management club of BIT Jaipur Campus that focuses on developing students' knowledge, skills, and employability in the fields of **Finance, Marketing, and Human Resource Management**. The club provides a platform for practical learning, industry interaction, analytical thinking, and professional development through various management-oriented activities and events.

Faculty Mentors

- **Dr. Roopali Sharma** – Professor, Department of Management
- **Dr. J. K. Chandel** – Assistant Professor, Department of Management

About the Club

The club aims to bridge the gap between classroom learning and industry requirements by exposing students to real-world business challenges and management practices. It brings together key managerial domains under one umbrella, including:

- Finance
- Banking
- Financial Markets
- Insurance
- Equities and Derivatives
- Marketing
- Digital Marketing
- Retail Management
- Branding and Advertising
- Human Resource Management (HR)
- Organizational Behaviour (OB)

The club encourages innovation, creativity, leadership, teamwork, and analytical thinking while preparing students for successful careers in the corporate world.

Major Activities

Finance & Business Activities

- **Budget Eye** – Analysis and discussion of the Union Budget.
- **Market Watch** – Business news analysis and discussion of current economic and management issues.
- **Stock Market Simulation Game** – Mock stock trading exercises based on market analysis and investment decisions.

Management Learning Activities

- **Case Folio** – Analysis and discussion of contemporary business cases and management challenges.
- **Paper Writing and Presentations**
- **Business Quizzes**
- **Group Discussions on Current Topics**
- **Role Playing Exercises** for developing analytical and decision-making skills.

Marketing & Creativity Activities

- **Ad Mad Competition** – Enhancing promotional and creative skills.
- **Brand Persona Development**
- **Branding and Advertising Exercises**

Professional Development Activities

- Workshops and Practical Training Sessions
- Guest Lectures by Industry Experts
- Interaction with Eminent Academicians and Corporate Professionals
- Team Building Activities

- Industry-Oriented Research and Development Projects

Objectives

- Enhance managerial and leadership skills.
- Develop analytical and decision-making abilities.
- Improve employability and industry readiness.
- Promote practical application of management concepts.
- Foster teamwork, communication, and professional competence.
- Provide exposure to contemporary business practices and industry trends.

Benefits of Joining the Club

Students associated with Galvanizers Club gain:

- Practical exposure to finance, marketing, and HR domains.
- Better understanding of business and corporate environments.
- Improved communication, leadership, and presentation skills.
- Networking opportunities with industry professionals.
- Enhanced career prospects and employability.
- Experience in problem-solving, business analysis, and strategic thinking.

The **Galvanizers Club** serves as a platform for aspiring managers to develop business acumen, leadership qualities, and professional skills necessary to excel in today's dynamic corporate environment.

9. **Multimedia Club**

The **Multimedia Club** at BIT Jaipur Campus is a creative and technology-driven student club that promotes learning and innovation in the fields of **Animation, Multimedia, Film Making, Game Design, Graphic Design, Visual Effects (VFX), and Digital Media**. The club provides students with opportunities to develop artistic, technical, and storytelling skills through practical projects, workshops, competitions, and industry interactions.

Major Technical Clubs

1. Animated Movies Club

- Focuses on animation film production and storytelling.
- Provides exposure to 2D and 3D movie-making techniques.
- Organizes film-making workshops, seminars, movie showcases, and animation competitions.
- Encourages individual and group animation film projects.

2. Animated Games Club

- Focuses on game design and development.
- Covers game programming, game art, level design, and game testing.
- Conducts gaming competitions, workshops, guest lectures, and game development activities.

- Introduces students to modern game development technologies and industry practices.

Key Areas of Learning

- 2D & 3D Animation
- Character Design and Storyboarding
- Graphic Design
- Visual Effects (VFX)
- Film Making and Editing
- Game Design and Development
- Digital Illustration
- Audio-Video Production
- Multimedia Content Creation

Activities

- Animation and game development projects
- Workshops and expert lectures
- Film screenings and highlights.
- Creative competitions
- Industry visits and studio tours
- Participation in cultural and technical festivals
- Student project exhibitions and portfolio development

Benefits

- Hands-on experience with industry-standard software and tools.
- Development of creativity, design thinking, and storytelling skills.
- Exposure to animation, gaming, and media industries.
- Opportunities to build professional portfolios and industry-ready skills.
- Enhanced career prospects in animation, multimedia, gaming, film production, graphic design, and digital content creation.

The Multimedia Club serves as a platform for students to combine **creativity, technology, and innovation** while preparing for careers in the rapidly growing animation, gaming, and digital media industries.

10. **Science Club**

The **Science Club – MaPhyChem**, established in **2023** by the Department of Sciences at BIT Jaipur Campus, is a multidisciplinary technical club that promotes scientific learning, research, innovation, and practical applications of concepts from **Mathematics, Physics, Chemistry, Electronics, and Environmental Science**. The club provides students with opportunities to explore emerging technologies, undertake research-oriented activities, and develop critical thinking skills through firsthand learning.

Club Mentor

Dr. Anand Kumar Srivastava

Associate Professor, Department of Sciences

Domains of the Club

1. Mathematical Theory

Core Discipline: Mathematics

Mentors:

- Dr. Gurminder Singh Malhotra
- Dr. Indra Kumar Dadheech
- Dr. Bhawna Agarwal

Objective:

The Mathematical Theory domain aims to strengthen students' understanding of fundamental mathematical concepts and bridge gaps in their foundational knowledge. It helps students develop analytical thinking and mathematical reasoning skills that are essential across engineering and science disciplines.

Activities:

- Concept clarification sessions
- Mathematical problem-solving discussions
- Theory enrichment workshops
- Analytical learning exercises

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2. Plasma Technology

Core Discipline: Physics

Mentor: Dr. Anand Kumar Srivastava

Co-Mentor: Dr. Ravindra Kumar

About the Domain:

Plasma, known as the **fourth state of matter**, is an ionized gas consisting of electrons, ions, and neutral particles. It has applications in advanced technologies ranging from semiconductor manufacturing to biomedical engineering.

Application Areas:

- Semiconductor fabrication and plasma etching
- Surface modification and material processing
- Plasma welding and cutting
- Plasma display technologies
- Air and water purification
- Biomedical applications
- Plasma nanotechnology
- Environmental pollution control

Activities:

- Learning through technical journals and magazines
- Plasma device design and modelling
- Workshops and practical training sessions
- Research paper writing and presentations.
- Expert lectures and discussions
- Industrial and institutional visits
- Research and development projects.

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3. Antenna Research

Core Discipline: Electronics & Communication Engineering and Electrical & Electronics Engineering

Mentor: Dr. Dheeraj Bhardwaj

About the Domain:

This domain focuses on the design, simulation, and development of modern antenna systems used in wireless communication technologies. It provides exposure to innovative research in RF and microwave engineering.

Key Areas:

- Microstrip Antennas
- Millimeter Wave Antennas
- PIFA Antennas
- Array Antennas
- IoT Communication Systems
- Wi-Fi, WLAN, LTE, 4G & 5G Technologies
- Biomedical Antenna Applications

Tools and Technologies:

- IE3D
- Ansys HFSS

Benefits:

- Research publication opportunities.
- Exposure to telecom and wireless industries
- Career preparation for organizations such as ISRO, DRDO, Qualcomm, Amazon, Meta, Bharat Electronics, and other leading technology companies

Activities:

- Antenna design and simulation
- Research projects
- Technical workshops
- Journal and conference publications.
- Expert lectures and seminars

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4. Green Environment

Core Discipline: Chemistry
Mentor: Dr. Ranjana Agrawal
Co-Mentors:

- Mrs. Nidhi Singh
- Dr. Rajni Agrawal

About the Domain:
The Green Environment domain promotes environmental awareness, sustainability, and responsible use of natural resources. It encourages students to actively participate in environmental conservation initiatives and develop eco-friendly practices.
Objectives:

- Promote environmental awareness among students.
- Develop environmental ethics and responsibility.
- Encourage conservation of natural resources.
- Support sustainable development initiatives.
- Inspire constructive action towards environmental protection.

Focus Areas:

- Environmental conservation
- Sustainability practices
- Green technologies
- Resource management
- Community awareness programs

Key Objectives of MaPhyChem Club

- Promote scientific curiosity and research culture.
- Bridge theoretical knowledge with practical applications.
- Encourage interdisciplinary learning across science and engineering.
- Develop analytical, technical, and critical thinking skills.
- Foster innovation through projects, workshops, and research activities.
- Enhance employability and higher education opportunities.
- Create awareness about emerging technologies and environmental sustainability.

Benefits of Joining the Club

- Firsthand exposure to advanced scientific and technological domains.
- Research and publication opportunities.
- Interaction with industry experts and academicians.
- Participation in workshops, seminars, and technical events.
- Development of analytical, technical, and innovation skills.
- Better preparation for careers in research, industry, and higher studies.

The **Science Club – MaPhyChem** serves as a platform for students to explore the fascinating world of Mathematics, Physics, Chemistry, and emerging

technologies while contributing to scientific advancement and sustainable development.